4. Assignment Topics

Choose the topic assigned to you. Read the scenario, then work out the happy path and all failure paths yourself.

Assignment 1: Student Project Approval & Allocation System

Scenario

A department runs a final-year project programme. Every student team must submit a project proposal. The proposal is screened, reviewed by a committee, and, once approved, a faculty guide is allocated. The process ends when the guide formally accepts the team and the student is notified.

Participants (lanes / pools)

- Student / Team
- Project Coordinator
- Review Committee (incl. HoD)
- Faculty Guide
- Project Management System

Process boundaries

- **Starts when:** A student team decides to submit a project proposal.
- **Ends when:** A guide is allocated and confirmed, or the proposal is closed as rejected / lapsed.

Your tasks

1. Draw the happy path from start to successful completion.
2. List every failure path you can find. Think about each step and ask what could go wrong.
3. For each failure, design a solution inside the diagram and record it in the Failure Path Register.
4. Choose task types and gateways deliberately, and be ready to justify each choice.

Where to look for failures

- Data and validation problems (forms, team rules, duplicate topics)
- Deadlines and review delays
- Committee decisions other than "approve"
- Resource problems (guide availability and workload)
- System or notification failures
- Withdrawal by the student or guide after approval

Assignment 2: Logistics & Shipment Exception Management

Scenario

A courier company moves parcels from a shipper to a recipient through pickup, hubs and last-mile delivery. Most shipments arrive without trouble, but many things can go wrong on the way. Model the full shipment life cycle with emphasis on how exceptions are detected and resolved.

Participants (lanes / pools)

- Shipper / Customer
- Control Tower / Dispatcher
- Pickup Agent
- Hub / Warehouse
- Delivery Agent
- Recipient
- Claims Department

Process boundaries

- **Starts when:** A shipper books a shipment.
- **Ends when:** The shipment is delivered and closed, returned to sender, or closed as lost or cancelled.

Your tasks

1. Draw the happy path from start to successful completion.
2. List every failure path you can find. Think about each step and ask what could go wrong.
3. For each failure, design a solution inside the diagram and record it in the Failure Path Register.
4. Choose task types and gateways deliberately, and be ready to justify each choice.

Where to look for failures

- Pickup and address problems
- Damage, loss and missing scans
- Delays caused by weather, breakdown or capacity
- Customs and documentation (for international shipments)
- Recipient unavailable, refusal, or cash-on-delivery failure
- Tracking or scanning system failures, and cancellation after dispatch

Assignment 3: Loan Origination & Approval System

Scenario

A bank handles personal and secured loan applications. An application passes through document checks, identity verification, credit assessment, underwriting, offer and acceptance, and finally disbursement. The model must show both the normal approval flow and every point at which an application can stall, be rejected or be cancelled.

Participants (lanes / pools)

- Applicant
- Loan Officer
- Underwriter / Credit Committee
- Fraud & Compliance Team
- Credit Bureau (external pool)
- Operations / Disbursement
- Core Banking System

Process boundaries

- **Starts when:** An applicant submits a loan application with supporting documents.
- **Ends when:** The loan is disbursed, or the application is rejected, withdrawn or has lapsed.

Your tasks

1. Draw the happy path from start to successful completion.
2. List every failure path you can find. Think about each step and ask what could go wrong.
3. For each failure, design a solution inside the diagram and record it in the Failure Path Register.
4. Choose task types and gateways deliberately, and be ready to justify each choice.

Where to look for failures

- Document and identity problems
- Fraud, AML or sanction flags
- Unavailable external services (credit bureau)
- Credit, income and collateral outcomes: reject, counter-offer or escalate
- Timeouts (offer validity, signature), withdrawal and refusal
- Disbursement failures

5. Evaluation Rubric (20 Marks)

Criterion	What is assessed	Marks
Happy path	Complete, correctly ordered, and traceable from start to successful end.	4
Task types	Each task uses the right type, with a clear reason.	3
Gateways	Correct use of exclusive, parallel and inclusive / event-based gateways, with proper splits and joins.	3
Failure identification	Coverage of realistic failure points across the whole process, not only the obvious ones.	4
Solutions	Each failure has a sensible resolution using proper constructs (boundary events, loops with limits, escalation, compensation).	4
Notation and readability	Valid BPMN 2.0, clear labels, lanes, consistent layout, and a good use of sub-processes.	2
Total		20

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6. Reference Solution Outline

Use this as a marking guide. Students may find valid failures and solutions that are not listed here; give credit when the resolution is sensible and shown in the model.

Assignment 1: Student Project Approval & Allocation System

Happy path

1. Student team submits the project proposal (user task).
2. System validates form completeness and team rules (service task).
3. Coordinator screens the proposal (user task).
4. System runs a similarity / duplicate-topic check (service task).
5. Review committee evaluates the proposal (user task), then an exclusive gateway decides: approved / revise / rejected.
6. On approval, a business rule task checks guide availability and workload against domain match.
7. System allocates a guide (service task) and sends an allocation request (send task).
8. Guide accepts the request (receive task / user task).
9. System notifies the student and records the allocation (send task + service task).
10. End event: project allocated.

Failure paths and solutions

ID	Failure point	Cause / trigger	Solution and BPMN construct
F1	Proposal validation	Missing fields, no abstract, invalid roll numbers.	Exclusive gateway after the validation task. Return the proposal with an error list; student resubmits. A loop counter limits this to 2 attempts, after which the case is escalated to the coordinator.
F2	Team rule check	Team size above the limit, or a student already belongs to another team.	Reject at validation, notify the student, and let the team be reformed. Ends with an error end event "Invalid team".
F3	Duplicate / plagiarised topic	Similarity score above the threshold against earlier projects.	Coordinator reviews the match. Either request a modified topic (loop back to the proposal task) or reject. Both outcomes are recorded.
F4	Proposal deadline	Team does not submit before the deadline.	Timer boundary event on the submission task. Flag the team and notify them. A late-submission request goes to the coordinator for approval.
F5	Committee asks for revision	Scope too large, objective vague, or novelty is weak.	Revision loop with a 7-day timer. Maximum 2 cycles, then the committee must approve or reject.
F6	Committee rejects	Not feasible, out of syllabus, or lacks resources.	Send reasons to the student. One new topic submission is allowed (loop to start); otherwise end "Rejected".
F7	Review delayed	Committee does not decide within the SLA.	Non-interrupting timer boundary event sends a reminder; a second timer raises an escalation event to the HoD.
F8	No guide available	Domain mismatch, or all matching guides are at maximum load.	Business rule returns "none". Coordinator allocates manually, appoints a co-guide or external guide, or places the team on a waiting list.

F9	Guide declines or is silent	Workload, conflict of interest, or no response.	Timer plus decline path. Exclude that guide and rerun the allocation rule. After 3 attempts the coordinator allocates manually.
F10	Notification / system failure	Email service down or database write error.	Error boundary event on the service task. Retry up to 3 times, then fall back to manual notification by the coordinator.
F11	Withdrawal after approval	Team dissolves or guide goes on leave.	Message boundary event on the allocation sub-process. Compensation releases the guide slot and returns the team to the allocation step.

Assignment 2: Logistics & Shipment Exception Management

Happy path

1. Shipper creates the shipment booking (user task).
2. System validates address, weight and service level, and calculates charges (business rule task).
3. Pickup is scheduled and a driver is assigned (service task).
4. Pickup agent collects the parcel and scans it (user / manual task).
5. A parallel gateway runs the physical movement and the tracking update with shipper notification (send task) together; they join afterwards.
6. Hub sorting and scanning, then line-haul to the destination hub (sub-process, multi-instance per leg).
7. Parcel goes out for delivery and the recipient is notified (send task).
8. Delivery agent hands over the parcel and captures proof of delivery (user task).
9. System closes the shipment and generates the invoice (service task).
10. End event: shipment delivered.

Failure paths and solutions

ID	Failure point	Cause / trigger	Solution and BPMN construct
E1	Pickup	Shipper absent or parcel not ready.	Error / escalation boundary event on the pickup task. Reschedule (max 2 attempts), then cancel the booking and charge a cancellation fee.
E2	Address validation / delivery	Incomplete or wrong address found at booking or at the doorstep.	At booking: return to shipper to correct. At delivery: agent contacts the recipient (message event). An address change may add a surcharge, decided at an exclusive gateway.
E3	Damaged parcel	Damage found at hub scan or on delivery.	Inspection task, then an exclusive gateway: is it insured? If yes, a claims sub-process leads to refund, replacement or reshipment. If not, record and inform the shipper.
E4	Lost parcel	No scan event for 48 hours (timer).	Tracing sub-process. If the parcel is not found within 5 days, declare it lost, pay compensation, notify the customer, and end "Lost".
E5	Delay in transit	Weather, strike, vehicle breakdown or hub overload.	Event-based gateway waits for either a delay message or a timer. Notify the customer, reroute or assign an alternate vehicle, and update the ETA.
E6	Customs hold	Missing or incorrect export/import documents.	Request documents from the shipper (send and receive tasks) with a 5-day timer. If provided, continue; otherwise return to origin.

E7	Recipient unavailable	No one present at the delivery address.	Loop with a counter for up to 3 delivery attempts. After that, hold at the depot for 7 days, then return to sender.
E8	Refusal / COD failure	Recipient refuses the parcel or cash-on-delivery payment fails.	Exclusive gateway on delivery outcome. Initiate return to sender and adjust charges.
E9	Tracking / scan failure	Scanner offline or data mismatch between systems.	Error boundary event on the scan task. Hub supervisor updates the record manually, then a reconciliation task runs.
E10	Cancellation after dispatch	Shipper cancels while the parcel is in transit.	Message boundary event on the transit sub-process. Stop movement, return the parcel, and run compensation (refund minus fee).

Assignment 3: Loan Origination & Approval System

Happy path

1. Applicant submits the application and documents (message start event / user task).
2. System checks document completeness (business rule task).
3. A parallel gateway runs KYC verification (service task), credit bureau score retrieval (service task via the external pool), and income / employment verification (user task) together, then joins.
4. Eligibility and risk scoring (business rule task); an exclusive gateway routes to auto-approve, manual review, or reject.
5. Underwriter reviews the case and approves (user task).
6. Loan offer is generated (script / service task) and sent to the applicant (send task).
7. Applicant accepts the offer (receive task).
8. Loan agreement is e-signed (user task).
9. Funds are disbursed to the verified account (service task), the loan account is created, and the applicant is notified.
10. End event: loan disbursed.

Failure paths and solutions

ID	Failure point	Cause / trigger	Solution and BPMN construct
L1	Document check	Incomplete or illegible documents.	Return to the applicant with a list of missing items and a 7-day timer; loop on resubmission. If there is no response, end "Application lapsed".
L2	KYC verification	Identity details do not match official records.	Exclusive gateway to a manual verification task by the loan officer. If unresolved, reject with a reason code.
L3	Fraud / AML screening	Forged documents, or an AML or sanctions-list hit.	Escalation event to the Fraud & Compliance pool; investigation sub-process. If confirmed, reject, file the internal report and end with a terminate event.
L4	Credit bureau call	API timeout or bureau down.	Error boundary event: retry 3 times at set intervals. Fallback is manual credit check by the officer, or park the application with a timer.
L5	Credit score / risk	Score below cutoff.	Exclusive gateway: reject with reasons, or counter-offer with a guarantor, collateral or smaller amount.
L6	Income / DTI	Insufficient income or debt-to-income ratio too high.	Offer a reduced amount or longer tenure and loop back to re-scoring (once). Otherwise reject.

L7	High-value / borderline case	Amount above the underwriter's authority, or a borderline score.	Escalate to the senior credit committee with an SLA timer. An inclusive gateway adds extra checks where the policy requires them.
L8	Collateral valuation	Valuation lower than expected, or title issues (secured loans).	Revalue, reduce the loan-to-value amount, and re-underwrite. If title is unclear, reject.
L9	Offer acceptance	Applicant does not respond within the offer validity (for example 15 days).	Timer boundary event on the receive task. Offer expires; end "Offer expired".
L10	Withdrawal at any stage	Applicant cancels the application.	Message boundary event on the main process. Compensation reverses processing fees and releases any holds. End "Withdrawn".
L11	E-signature	Agreement not signed, or signature mismatch.	Resend and remind. After 2 failed attempts, cancel the offer and notify the applicant.
L12	Disbursement	Invalid bank account or payment gateway error.	Account verification step, then retry. If it still fails, hold funds and contact the applicant. Compensation reverses the loan account creation.